

# Lecture 9b: Nested Cross-Validation

Brain-Computer Interfacing (BKI 323)

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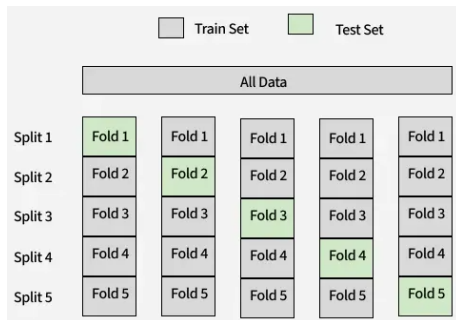
# Standard cross-validation

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- Model evaluation: **estimating** a **generalization** performance
- Note: cross-validation **does not prevent overfitting!**



Please note cross-validation schemes: leave-one-out, k-fold, stratified, etc.

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- **Test** data: evaluate the learned model

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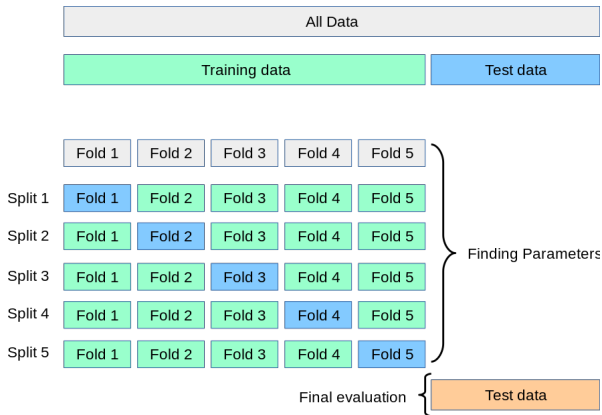
- **Train** data: learn the model parameters
- **Test** data: evaluate the learned model

What about **hyper-parameters**? 🙌🙌

- What are hyper-parameters?
- How are they learned?
- What may happen if we learn them using a standard k-fold cross-validation, evaluating the effect of hyper-parameters using the test data?

# Nested cross-validation

We need separate data to **fit** the **hyper-parameter(s)**, so we **split** the **train** data into **train** and **validation** data.



# Nested-cross-validation pseudo-code

```
for outerfold in outerfolds:
    split data into train and test

    for hyperparam in hyperparams:

        for innerfold in innerfolds:
            split train into train_ and validation

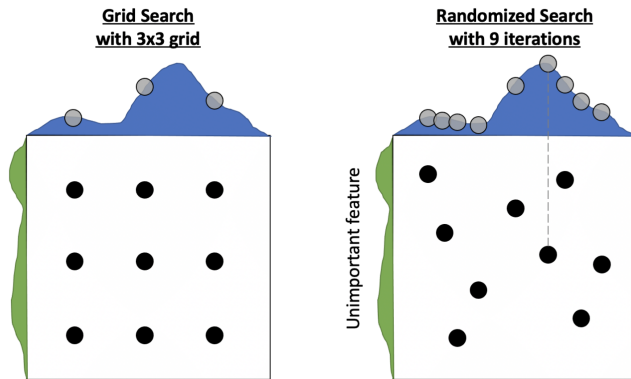
            model.fit(train_, hyperparam)
            model.predict(validation)

        choose best hyperparam

    model.fit(train, besthyperparam)
    model.predict(test)
```



# Grid search and Randomized search



What could be a downside of a grid/randomized search? 🙌🙌